Benchmarking Dynamic Frequency Estimators for Low-Inertia IBR Grids: A Latency–Robustness Trade-off Analysis

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***Abstract*—In low-inertia grids dominated by Inverter-Based Resources (IBRs), the latency and robustness of frequency estimation are critical for protection and dynamic security assessment. This paper evaluates twelve estimators from five algorithmic families under five stress scenarios using dual-rate numerical simulation (1 MHz physics / 10 kHz DSP). Each estimator is independently tuned by scenario-wise grid search and assessed on RMSE, peak error, trip-risk duration (T_{trip}), and computational cost. Tests (A)–(C) follow the IEC/IEEE 60255-118-1 test philosophy; Scenarios (D)–(E) introduce IBR-specific phase discontinuities and harmonic distortion beyond the standard scope. Under a sustained frequency ramp ($+3$ Hz/s), the RoCoF-Augmented Extended Kalman Filter (RA-EKF) achieves RMSE $= 0.00823$ Hz—$8.15\times$ lower than SRF-PLL. Under composite islanding, the RA-EKF incurs zero threshold breaches across all $N = 30$ Monte Carlo trials ($T_{\mathrm{trip}} = 0$), versus $T_{\mathrm{trip}} = 0.120$ s for the standard EKF. Window-based and loop-based methods exhibit complementary strengths across scenario types. In the evaluated scenarios, data-driven approaches exhibited higher computational cost and did not demonstrate consistent accuracy gains over scenario-optimized classical methods. The findings indicate that several estimators passing IEC/IEEE 60255-118-1 standard tests degrade under composite disturbances, suggesting that the standard test scope may not fully characterize estimator behavior in IBR-dominated conditions.**

***Index Terms*—Frequency Estimation, Inverter-Based Resources (IBR), Low-Inertia Systems, Extended Kalman Filter (EKF), Phase-Locked Loop (PLL), RoCoF Measurement, Sub-Cycle Transient Tracking, Taylor-Fourier Methods, Benchmarking, Protection and False-Trip Risk.**

I. Introduction

The transition toward inverter-dominated power systems is reducing grid inertia and accelerating frequency dynamics. Frequency deviations and Rate of Change of Frequency (RoCoF) can exceed protection thresholds within milliseconds [1], [2]. Unlike synchronous machines, Inverter-Based Resources (IBRs) regulate power electronically, decoupling the primary energy source from electrical frequency and increasing the impact of harmonics, phase discontinuities, and nonstationary disturbances on protection and dynamic security assessment [1], [3], [4]. In IBR-dominated grids, phase discontinuities can induce cycle-slips in phase-locked loops, sustained RoCoF can exceed the tracking capacity of multi-cycle window methods, and converter-driven harmonic injection can cause spectral leakage in DFT-based estimators. These failure modes may occur simultaneously, yet existing benchmarks typically assess them in isolation [5], leaving a gap for protection-class applications.

The main estimator families address the latency–robustness trade-off differently. SRF-PLL is widely used but susceptible to cycle-slip under abrupt phase discontinuities [6], [7]. Window-based methods (IpDFT, TFT) provide strong harmonic rejection at the cost of structural observation latency [8], [9]. Model-based filters (EKF, UKF) improve tracking under transients [10], [11], while data-driven approaches may offer accuracy gains at higher computational cost [5], [12].

This paper presents a stress-oriented benchmark of twelve estimators from five algorithmic families, evaluated in a local, non-time-synchronized relay-class configuration under five scenarios: (A)–(C) follow the IEC/IEEE 60255-118-1 test philosophy; (D)–(E) introduce IBR-specific disturbances beyond the standard scope. The main contributions are: (1) a Composite Islanding scenario that exposes cycle-slip failure modes; (2) scenario-wise grid-search tuning with dual-rate simulation (1 MHz physics / 10 kHz DSP); and (3) a comparative analysis of accuracy, settling time, trip-risk duration, and computational cost. Within this benchmark, the adapted RA-EKF augments the EKF state vector with an explicit $\dot{\omega}$ term and adds event-gating to reduce false-trip duration under phase discontinuities at moderate computational cost.

II. Related Work and Algorithmic Landscape

Frequency estimation in low-inertia grids involves a fundamental latency–robustness trade-off [5]. Window-based methods (IpDFT, TFT) provide strong harmonic rejection but impose structural observation latency; IpDFT assumes a stationary phasor over its window, whereas TFT employs a Taylor-series model to improve RoCoF tracking at the cost of higher noise sensitivity [8], [9]. Loop-based methods [SRF-PLL, Second-Order Generalized Integrator Frequency-Locked Loop (SOGI-FLL)] are computationally efficient but susceptible to cycle-slip under abrupt phase discontinuities [13]. Adaptive recursive estimators [Recursive Least Squares (RLS), Variable

Forgetting Factor RLS (VFF-RLS)] fit an AR(2) signal model and may diverge under severe transients. Model-based filters (EKF, UKF, RA-EKF) offer an improved latency–robustness balance [10], [11]; the RA-EKF adds Innovation-Driven Covariance Scaling and Event Gating to address IBR-specific islanding transients. Data-driven approaches (Koopman, PI-GRU) achieve competitive accuracy in some regimes but differ substantially in computational cost [5], [12]. Prior studies typically use fixed tuning parameters and evaluate disturbances in isolation; combined phase discontinuities, sustained RoCoF, and harmonic distortion are not assessed simultaneously. Table I provides a qualitative comparison of eleven of the twelve evaluated estimators.

III. Methodology and Test Framework

A dual-rate simulation framework separates continuous-time system physics ($f_{\mathrm{s,phys}} = 1\,\mathrm{MHz}$) from discrete-time estimation ($f_{\mathrm{s,DSP}} = 10\,\mathrm{kHz}$). This yields negligible numerical error in ground-truth frequency trajectories while reproducing the sampling constraints of commercial protection relays.

A. Evaluated Estimators

Twelve estimators from five algorithmic families are evaluated (Table I). **Loop-based:** SRF-PLL and SOGI-FLL with PI gains, IIR bandpass pre-filter, and FastRMS adaptive amplitude normalizer (not a standard textbook SOGI-FLL [13]). A unit inconsistency in the SOGI-FLL adaptation law (rad/s vs. Hz, factor 120π rad/s) was identified and corrected prior to evaluation. The narrow effective passband created by the bandpass pre-filter and adaptive normalizer suppresses apparent frequency deviation under a phase jump but cannot track sustained linear RoCoF, explaining the divergent ranking across Scenarios B and D. **Window-based:** IpDFT with a Hann window and three-point magnitude-parabolic interpolation [14] (the complex Jacobsen estimator is unsuitable here because Hann-window sidelobes are negative-real, which inverts the interpolation direction); TFT with a 2nd-order Taylor-series expansion of the fundamental component, Hann-weighted, which improves RoCoF tracking relative to a static DFT at the cost of higher noise sensitivity [9]. **Adaptive recursive:** RLS and VFF-RLS with a 2nd-order AR(2) bandpass pre-whitening filter. **Data-driven:** Koopman (RK-DPMU) and PI-GRU. **Model-based:** EKF, UKF, and RA-EKF. The column "RoCoF Out." indicates whether RoCoF is available as an explicit output (S) or derived by implicit differentiation (D). The UKF includes a 10-sample output moving average not present in the classic EKF.

1) Implemented RA-EKF Estimator

RoCoF state augmentation in the EKF has been studied previously [10], [11]. The RA-EKF uses a 4-state vector $[\theta_k, \omega_k, A_k, \dot{\omega}_k]^\top$ with second-order kinematic prediction ($\theta_k = \theta_{k-1} + \omega_{k-1}\Delta t + \frac{1}{2}\dot{\omega}_{k-1}\Delta t^2$; $\omega_k = \omega_{k-1} + \dot{\omega}_{k-1}\Delta t$) and two IBR-tailored mechanisms. *Innovation-Driven Covariance Scaling:* $\mathbf{Q}_k = \mathrm{clip}(r_k, 0.25, 4)\,\mathbf{Q}_{\mathrm{base}}$, where $r_k = |\nu_k|/\hat{\sigma}_\nu$, inflates process noise during transients and reduces it at steady state. *Event Gating:* when $|\nu_k| > \gamma\hat{\sigma}_\nu$ ($\gamma = 2.0$), a soft re-initialization updates θ_k only, preserving ω_k and $\dot{\omega}_k$ to limit cycle-slip behavior at phase discontinuities. This heuristic trades strict stochastic optimality for bounded nonlinear stability during abrupt phase steps.

B. Test Scenarios and Noise Model

Five scenarios stress the estimators under representative IBR-grid conditions (Fig. 1). Tests (A)–(C) use Additive White Gaussian Noise (AWGN) with $\sigma = 0.001\,\mathrm{pu}$; Scenario (D) uses $\sigma = 0.005\,\mathrm{pu}$; and Scenario (E) uses $\sigma = 0.003\,\mathrm{pu}$ base noise plus sparse impulsive samples (amplitude $\leq 0.05\,\mathrm{pu}$, probability $< 1\%$).

(A) Magnitude Step: $+10\%$ voltage step with zero phase angle at the disturbance onset.

(B) Frequency Ramp: $+3.0\,\mathrm{Hz/s}$ RoCoF (IEEE 1547-2018 Category III compliant).

(C) AM Modulation: 10% depth at 2 Hz.

(D) Composite Islanding: instantaneous $+60^\circ$ phase jump (maximum mismatch during breaker reclosing [7]), 4% 5th harmonic (250 Hz), 2% 7th harmonic (350 Hz), 0.5% 32.5 Hz interharmonic (a sub-synchronous component characteristic of IBR converter switching; Total Harmonic Distortion THD $\approx$ 4.5%, IEEE 519 compliant), and $\sigma = 0.005\,\mathrm{pu}$ noise.

(E) IBR Multi-Event: 5 s sequence with harmonic-rich steady state (5th: 4%, 7th: 2%; THD $\approx$ 4.5%, IEEE 519 compliant), a $+40^\circ$ phase jump, a $-3.0\,\mathrm{Hz/s}$ RoCoF segment reaching a nadir of $57.5\,\mathrm{Hz}$ (IEEE 1547-2018 UFLS limit), a $+80^\circ$ jump with second-order ring-down recovery, and impulsive noise [11].

Scenarios (A)–(C) follow the IEC/IEEE 60255-118-1 test philosophy, whereas Scenarios (D)–(E) introduce IBR-specific disturbances beyond the standard scope.

C. Performance Metrics and Optimization Protocol

A scenario-wise grid search over the hyperparameter ranges in Table II selects the configuration minimizing RMSE for each estimator–scenario pair, identifying the performance ceiling of each algorithmic family under each disturbance type independent of sub-optimal field parameterization. Let $e(t) = \hat{f}(t) - f_{\mathrm{true}}(t)$. Four metrics are reported: **RMSE**, **Peak Error** ($|e|_{\max}$), **Settling Time** (first time $|e|$ falls and remains below $0.2\,\mathrm{Hz}$), and **Trip-Risk Duration** (T_{trip}, cumulative time with $|e| > 0.5\,\mathrm{Hz}$).

Computational cost is the mean time per sample over 20 runs on an Intel Core Ultra 5 125H (single-threaded Python 3.13) and is used as a relative comparison; reported values are Python baselines, not C or FPGA projections.

For the non-normative compliance heatmap (Fig. 2g), a *Pass* requires RMSE $< 0.05\,\mathrm{Hz}$, $|e|_{\max} < 0.5\,\mathrm{Hz}$, and $T_{\mathrm{trip}} < 0.1\,\mathrm{s}$. These thresholds are drawn from IEC/IEEE 60255-118-1 accuracy classes and are used here for visualization only.

IV. Simulation Results and Benchmarking

A. Performance on Standard IEC/IEEE Scenarios

Table III summarizes performance on standard scenarios (3 s.f.). **Steady-state:** SRF-PLL achieves RMSE $= 1.30\times10^{-4}$ Hz

TABLE I: Unified Qualitative Benchmark of Frequency Estimators (relay-class, non-synchronized context). Scores: 0=Fail, 1=Poor, 2=Fair, 3=Excellent. h_{HU}: Harmonic/Unbalance Immunity; $r_{32.5}$: rejection of 32.5 Hz IBR interharmonic (a converter-switching-characteristic sub-synchronous component used in Scenarios D–E; see Section III-B); n_{sens}: noise floor sensitivity; n_{NG}: Non-Gaussian noise robustness; Latency class: P1 $<$ 20 ms, P2: 20–40 ms, M1 $>$ 40 ms, P3: sub-cycle ($<$1 ms); v_{RT}: real-time viability (3=embedded, 0=GPU only); RoCoF Out.: S = explicit state output, D = implicit differentiation; *Phase Resp.*: phase-jump settling (D=cycle-slip, S=settles); †IpDFT latency = N_c/f_0: 33 ms at $N_c = 2$ to 167 ms at $N_c = 10$.

Method	Phase Resp.	RoCoF Out.	h_{HU}	$r_{32.5}$	n_{sens}	n_{NG}	Latency Class	v_{RT}	Comp. Order	Dominant Failure Mode
I. SOTA Methods (Post-2020)										
VFF-RLS	5–20 ms	D	2	3	3	2	P2	3	$O(p^2)$	VFF blow-up
UKF	$<$20 ms	S	2	2	2	2	P1	1	$O(n^3)$	Sigma-pt collapse
Koopman (DPMU)	40–80 ms	S	2	2	2	1	M1	1	$O(n^2)$	Model mismatch
PI-GRU (DL)	5–15 ms	S	3	3	3	3	P1	0	$O(\mathrm{Conv})$	Dataset shift
II. Industrial / Advanced Digital										
RA-EKF (Prop.)	$<$20 ms	**S**	2	2	3	2	**P1**	1	$O(n^3)$	Jacobian divergence
TFT	$<$20 ms	S	2	2	2	1	P1	1	$O(N^2)$	Runge overshoot
SRF-PLL	20–40 ms	D	1	1	1	0	P2	3	$O(1)$	Cycle slip / Lag
SOGI-FLL	20–40 ms	D	2	1	2	1	P2	3	$O(1)$	Tuning trade-off
IpDFT†	$\geq$20 ms	D	3	1	3	0	M1	2	$O(N \log N)$	Leakage / Latency
III. Legacy Baselines										
RLS (fixed λ)	5–20 ms	D	1	1	1	0	P2	3	$O(p^2)$	Covariance blow-up
TKEO	$3T_s$	D	0	0	0	0	P3	3	$O(1)$	Noise amplification

(Modulation); window-based methods reach RMSE $< 3 \times 10^{-4}$ Hz. **Ramp:** The Ramp test exposes structural limitations. SOGI-FLL fails to track (RMSE = 1.12 Hz; type-1 loop). The RA-EKF achieves RMSE = $\mathbf{0.00823}$ Hz—8.15× lower than SRF-PLL and 1.41× lower than EKF.

B. Robustness Under Severe IBR Disturbances

1) Composite Islanding (Scenario D)

Window-based methods exhibit spectral leakage: IpDFT reaches RMSE = 0.484 Hz (peak 2.91 Hz); TFT peak 2.74 Hz. EKF/UKF, lacking phase-reset mechanisms, incur $T_{\mathrm{trip}} = 0.122$ s; SRF-PLL avoids leakage (RMSE = 0.0450 Hz, $T_{\mathrm{trip}} = 0$). The **RA-EKF** achieves RMSE = $\mathbf{0.0393}$ Hz—a $\mathbf{5.7}$× RMSE reduction vs. EKF—with zero threshold breaches ($T_{\mathrm{trip}} = 0$; peak 0.365 Hz < 0.5 Hz, Fig. 2g).

2) Multi-Event Stress Test (Scenario E)

Table IV quantifies trip-risk under the IBR Multi-Event sequence (5th/7th harmonics; THD ≈ 4.5%, IEEE 519 compliant). SOGI-FLL incurs the highest trip-risk (3.10 s); SRF-PLL accumulates $T_{\mathrm{trip}} = 0.353$ s (RMSE = 0.307 Hz). The **RA-EKF** achieves RMSE = 0.159 Hz with $T_{\mathrm{trip}} = 0.081$ s at 88.5 μs/sample—a $\mathbf{4.4}$× trip-risk reduction vs. SRF-PLL and $\mathbf{4.5}$× vs. EKF (event-gating). IpDFT: $T_{\mathrm{trip}} = 0.563$ s; Koopman: 0.068 s.

TABLE II: Tuning Parameter Search Space. K_p [pu/rad], K_i [pu/(rad·s)]; Q [rad^2/s^2 for ω, pu^2 for A]; R [pu^2]; $\hat{\sigma}_\nu$, λ [dimensionless]; γ: RA-EKF event-gating threshold (dimensionless); g_{FLL}: SOGI-FLL adaptation gain (dimensionless); N_c [cycles]; N_{smooth} [samples].

Method	**Parameter**	**Search Range**
SRF-PLL	K_p, K_i	$[1, 60]$, $[1, 200]$ (linear)
EKF / RA-EKF	Q	10^{-4}–10^{4} (log)
	R	10^{-5}–10^{0} (log; $R_{\max} = 1.0$)
RA-EKF only	$\hat{\sigma}_\nu$, γ	$\{0.1, 0.3\}$, 2.0 (fixed)
SOGI-FLL	k, g_{FLL}	$\{0.5\ldots2.0\}$, $\{5\ldots150\}$
IpDFT	N_c	$\{2, 3, 4, 6, 8, 10\}$ cycles
RLS / VFF	λ, N_{smooth}	0.9–0.99999, 2–2000 samp.
TFT	N_c	$\{2, 3, 4, 6\}$ cycles
TKEO	Smooth window	10–50 samples

C. Computational Feasibility

Fig. 2e and Table V show the cost–risk Pareto frontier. **Low cost ($\leq$30 μs):** SRF-PLL (16.7 μs) and EKF (29.5 μs) are the least expensive; SRF-PLL, however, incurs elevated trip-risk under the composite Multi-Event sequence ($T_{\mathrm{trip}} = 0.353$ s; Table IV). **Moderate (30–100 μs):** SOGI-FLL (41.7 μs), TFT (45.6 μs), and RA-EKF (88.5 μs) are plausible candidates for embedded DSP/FPGA relay platforms, pending C or hardware-in-the-loop validation. **Higher cost ($>$100 μs):** IpDFT (110 μs), Koopman (120 μs), UKF (215 μs), and PI-GRU (≈154 ms/sample) are included for reference; PI-GRU is prohibitive without hardware acceleration.

V. Conclusion

The simulation results indicate that standard IEC/IEEE 60255-118-1 evaluation tests may not fully characterize frequency estimator behavior in converter-dominated grids, where phase discontinuities, high RoCoF, and harmonic distortion can expose failure modes that isolated single-disturbance tests do not reveal. This benchmark evaluates twelve estimators across five algorithmic families under five stress scenarios and reports three main observations.

1) **Latency–Robustness Trade-off:** Window-based methods achieve strong steady-state accuracy, but their behavior differs across scenario types. IpDFT is better suited to metering-oriented operation, whereas TFT reduces observation delay at the cost of higher noise sensitivity under composite disturbances. Loop-based estimators (SRF-PLL, SOGI-FLL) are computationally efficient and handle isolated phase jumps well, but exhibit elevated trip-risk under the composite Multi-Event sequence ($T_{\mathrm{trip}} = 0.426$ s for

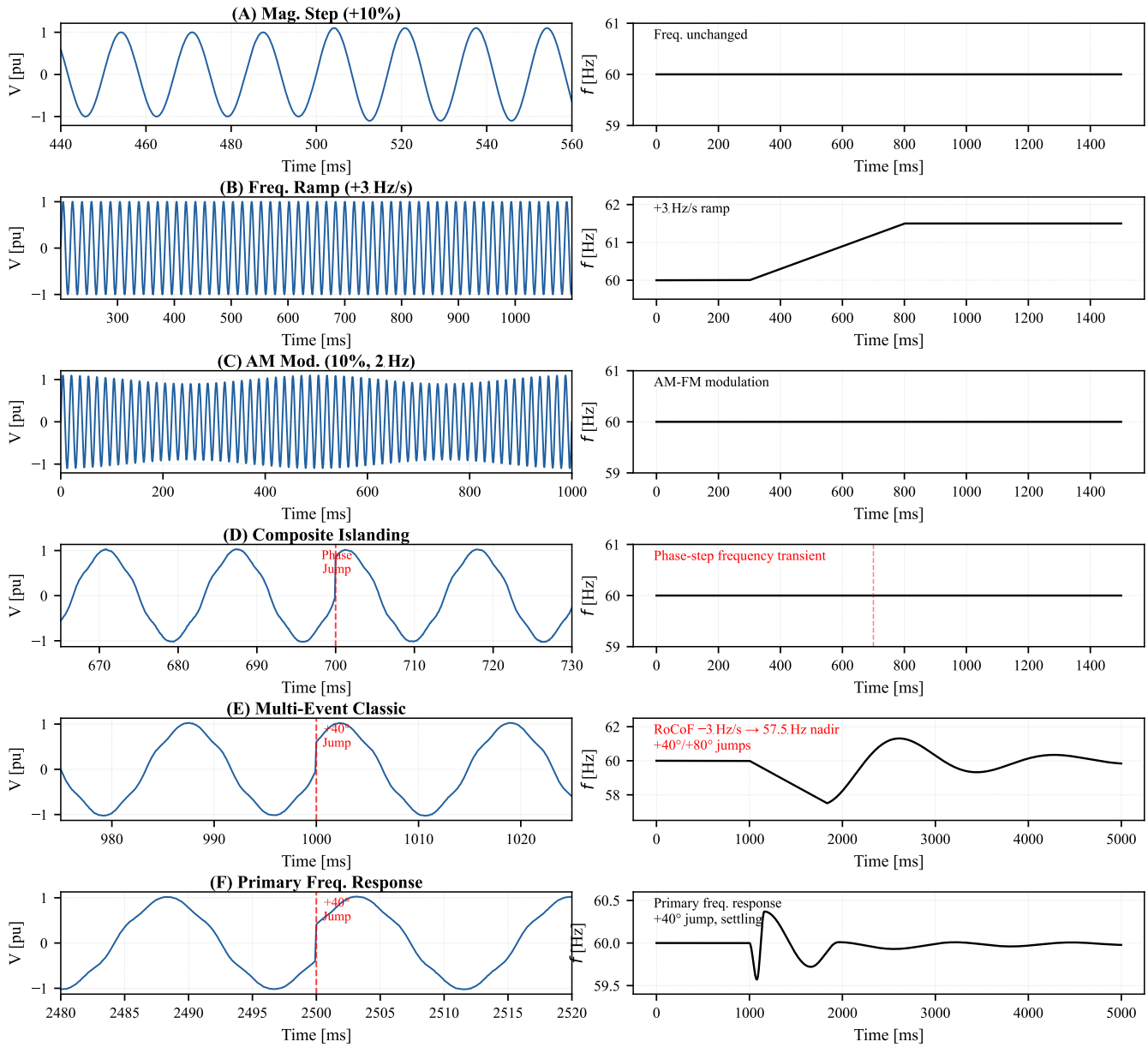

Fig. 1: Complete Stress-Test Scenario Suite. (A–C) Dynamic performance profiles following the test philosophy of IEC/IEEE 60255-118-1 (step, ramp, modulation). (D) Composite Islanding featuring an instantaneous $+60^\circ$ phase jump with harmonics and elevated noise ($\sigma = 0.005$ pu; Section III-B). (E) IBR Multi-Event: scenario-dependent noise ($\sigma_{\text{base}} = 0.003$ pu plus sparse impulsive spikes ≤ 0.05 pu); frequency profile shows the full 5 s composite sequence.

SRF-PLL).

2) **Model-Based Estimation:** Among the model-based estimators, the **RA-EKF**—combining an explicit $\dot{\omega}$ state with Innovation-Driven Covariance Scaling and Event Gating—achieves a favorable trade-off between tracking accuracy and trip-risk. Under Composite Islanding, the RA-EKF incurs zero threshold breaches across all $N = 30$ Monte Carlo trials ($T_{\text{trip}} = 0$), versus $T_{\text{trip}} = 0.120$ s for the standard EKF, at a computational cost of $68.9\,\mu$s/sample (Python baseline; C/HIL validation remains future work).
3) **Limits of Standard Compliance Tests:** Several estimators passing IEC/IEEE 60255-118-1 step and ramp tests exhibited degraded performance under composite multi-disturbance scenarios, suggesting that standard compliance tests may not fully capture estimator resilience in IBR-dominated conditions.

Data-driven approaches can be competitive in some scenarios, but computational cost varies considerably. PI-GRU requires approximately 145 ms/sample (Python/PyTorch CPU baseline), which is prohibitive for embedded real-time protection without hardware acceleration. In the evaluated scenarios, the pre-trained PI-GRU model did not outperform scenario-

TABLE III: Performance on Standard IEC/IEEE 60255-118-1 Scenarios [RMSE / Peak Error (Hz); mean $\pm$ std, Monte Carlo $N = 30$]. SOGI-FLL: corrected FLL adaptation law; Ramp result is outside loop bandwidth (type-1 loop, no steady-state RoCoF tracking). IpDFT: Hann window, Jacobsen complex spectral interpolation [14]; grid search selects $N_c = 2$ cycles for Ramp (33 ms) and $N_c = 10$ for Step/Mod (167 ms). TKEO, RLS, VFF-RLS, and PI-GRU excluded (divergence or off-scale error).

	Step (+10%) ($\mu \pm \sigma$)		**Ramp (+3 Hz/s)** ($\mu \pm \sigma$)		**Modulation** ($\mu \pm \sigma$)	
Method	**RMSE**	**Peak**	**RMSE**	**Peak**	**RMSE**	**Peak**
Window-based						
IpDFT	0.00160 ± 0.00004	0.00753 ± 0.00023	0.03525 ± 0.00002	0.08299 ± 0.00138	0.000260 ± 0.000020	0.000680 ± 0.000100
TFT	0.00212 ± 0.00003	0.01898 ± 0.00023	0.03071 ± 0.00002	0.05566 ± 0.00067	0.000260 ± 0.000020	0.000820 ± 0.000130
Data-driven						
Koopman	0.00125 ± 0.00004	0.00879 ± 0.00021	0.02856 ± 0.00002	0.05466 ± 0.00097	0.00123 ± 0.00002	0.00220 ± 0.00012
Loop-based						
SRF-PLL	0.000140 ± 0.000000	0.000730 ± 0.000000	0.06709 ± 0.00002	0.17260 ± 0.00028	0.000130 ± 0.000000	0.000250 ± 0.000000
SOGI-FLL	0.00899 ± 0.00000	0.01342 ± 0.00000	1.117 ± 0.00003	1.555 ± 0.00003	0.00899 ± 0.00000	0.01342 ± 0.00000
Model-based						
EKF	0.000240 ± 0.000010	0.00252 ± 0.00003	0.01160 ± 0.00011	0.04106 ± 0.00308	0.000150 ± 0.000000	0.000570 ± 0.000060
UKF	0.000280 ± 0.000040	0.00257 ± 0.00004	0.01273 ± 0.00016	0.04524 ± 0.00370	0.000280 ± 0.000020	0.00170 ± 0.00016
RA-EKF	0.00231 ± 0.00002	0.02988 ± 0.00024	0.00823 ± 0.00010	0.03178 ± 0.00246	0.00478 ± 0.00003	0.01278 ± 0.00041

optimized classical methods on RMSE or trip-risk, suggesting that a fixed pre-trained model cannot substitute for scenario-specific tuning under these disturbance conditions. This benchmark is entirely simulation-based; hardware-in-the-loop and physical relay validation remain as next steps.

The results suggest that the IEC/IEEE 60255-118-1 test scope may not fully characterize estimator resilience in IBR-dominated grids, as it lacks composite phase-step evaluations under simultaneous harmonic distortion. A *Composite Phase-Step with Harmonic Distortion* test—analogous to Scenarios D–E—evaluated via probabilistic compliance bounds ($\mathrm{PCB} = \mu_{|e|} + 3\sigma_{|e|}$) may provide a more representative assessment of relay security under realistic IBR conditions.

TABLE IV: Trip-Risk Analysis: Multi-Event Scenario (E). T_{trip}: cumulative time with $|e(t)| > 0.5$ Hz; CPU: mean time per sample (μs), Scenario E run. Values: mean $\pm$ std from Monte Carlo (N=30). Methods not shown: TKEO (RMSE $\geq$ 7.8 Hz, diverges); RLS (0.342 Hz, $T_{\mathrm{trip}} = 0.367$ s, CPU $= 12.7\,\mu$s; similar to EKF); RLS-VFF (0.874 Hz, $T_{\mathrm{trip}} = 2.20$ s); TFT ($T_{\mathrm{trip}} = 0.349$ s); PI-GRU (1.83 Hz, $T_{\mathrm{trip}} = 3.76$ s, CPU $\approx 1.54 \times 10^5\,\mu$s; Table V).

Method	**RMSE [Hz]** ($\mu \pm \sigma$)	T_{trip} **[s]** ($\mu \pm \sigma$)	**CPU [μs]**
RA-EKF	0.159 ± 0.003	0.081 ± 0.001	88.5
EKF	0.335 ± 0.000	0.364 ± 0.001	29.5
Koopman	0.471 ± 0.000	0.068 ± 0.000	120
UKF	0.337 ± 0.000	0.366 ± 0.001	215
SRF-PLL	0.307 ± 0.000	0.353 ± 0.001	16.7
IpDFT	0.636 ± 0.000	0.563 ± 0.000	110
SOGI-FLL	0.826 ± 0.000	3.100 ± 0.000	41.7

TABLE V: Computational Cost per Sample (Python Baselines). CPU: mean time per sample over 20 runs on an Intel Core Ultra 5 125H (single-threaded Python 3.13; not C or FPGA projections). Deployment class: P1 ($<20\,\mu$s, embedded MCU), P2 (20–40 μs, DSP), M1 ($>40\,\mu$s, DSP/FPGA or higher); these are per-sample execution time thresholds, not estimator response-latency classes.

Method	**Time/Sample** (μs)	**Class**	**Target HW**
SRF-PLL	$16.7\,\mu$s	P1	Low-end MCU
EKF	$29.5\,\mu$s	P2	DSP
SOGI-FLL	$41.7\,\mu$s	M1	MCU / DSP
TFT	$45.6\,\mu$s	M1	DSP
RA-EKF	$\mathbf{88.5}\,\mu$s	**M1**	**DSP / FPGA**
IpDFT	$110\,\mu$s	M1	DSP / FPGA
Koopman	$120\,\mu$s	M1	CPU
UKF	$215\,\mu$s	M1	High-perf CPU
PI-GRU	$153{,}900\,\mu$s ($\approx$154 ms)	—	GPU only

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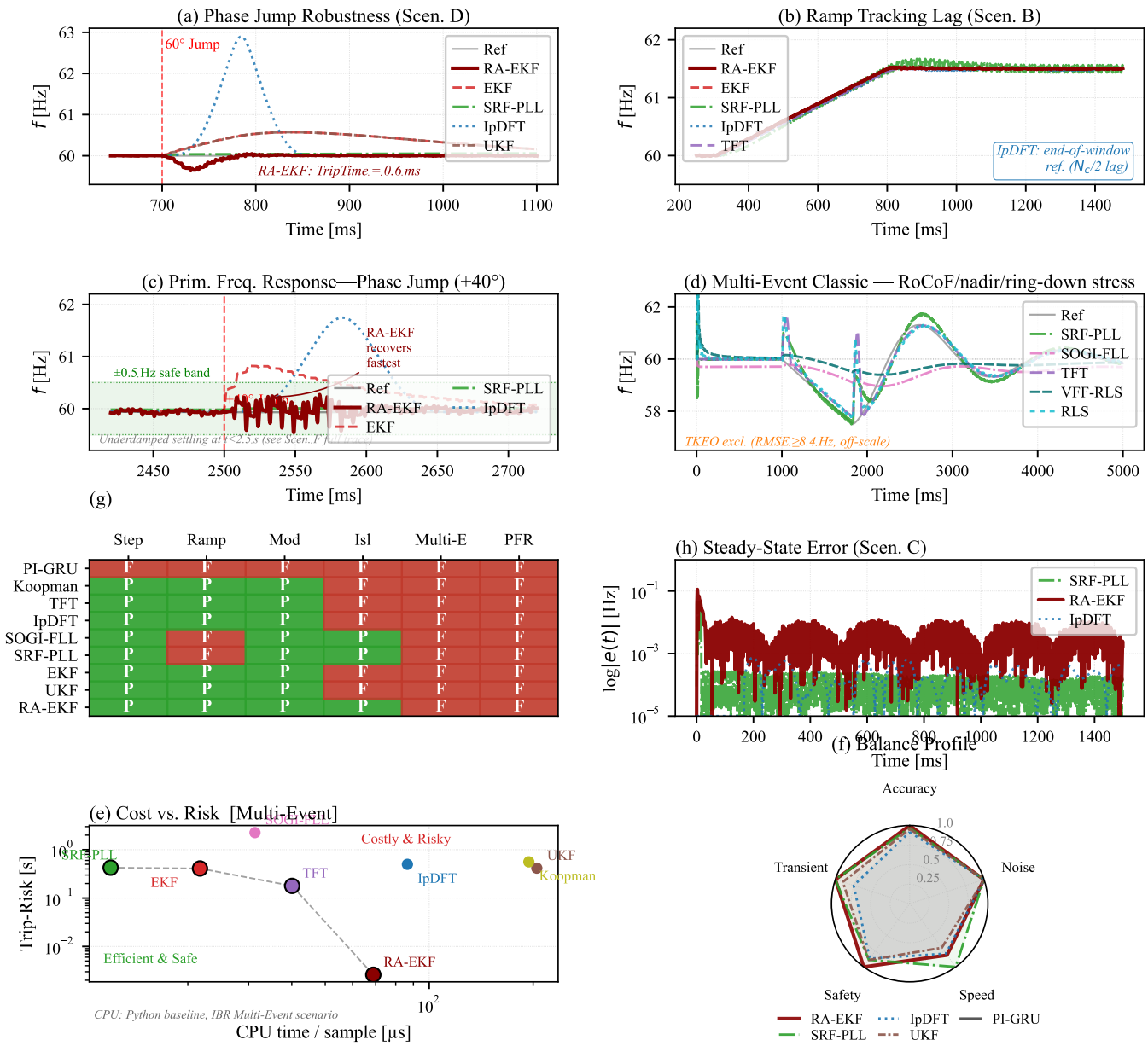

Fig. 2: Comprehensive Performance Dashboard. **(a)** Phase-Jump Robustness (Scen. D): RA-EKF event-gating limits trip-risk to ≈ 0.40 ms vs. 122 ms for standard EKF. **(b)** Ramp Tracking (Scen. B): RA-EKF achieves lowest RMSE via explicit $\dot{\omega}$ state; IpDFT uses 2-cycle grid-search window (33 ms) and achieves RMSE $= 0.035$ Hz; SOGI-FLL fails (RMSE $= 1.12$ Hz), outside loop bandwidth. **(c–d)** Multi-Event: stable vs. unstable estimators (TKEO excluded, RMSE ≥ 7.8 Hz). **(e)** Pareto frontier (†PI-GRU excluded, CPU ≈ 154 ms/sample, Trip ≈ 3.8 s). **(f)** Radar profile. **(g)** Compliance heatmap [RMSE < 0.05 Hz, Peak < 0.5 Hz, Trip < 0.1 s]; RA-EKF Islanding: peak $= 0.365$ Hz (below 0.5 Hz threshold; $T_{\text{trip}} = 0$). **(h)** Steady-state error (log scale, Scen. C). Subfigures: (a)–(d) time-domain, (e)–(f) aggregate, (g)–(h) compliance/steady-state.

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